Chapter 2. Topology of Euclidean Space

Open and closed sets, Bolzano–Weierstrass, Cauchy sequences, series, compactness, connectedness

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2.1 Euclidean space

$\mathbb{R}^n$ is the set of n-tuples $x = (x_1, \dots, x_n)$ with coordinatewise addition and scalar multiplication, the **inner product** $\langle x, y\rangle = \sum x_i y_i$, the **norm** $\|x\| = \langle x, x\rangle^{1/2}$ and the **distance** $\|x - y\|$.

Theorem 2.1 (2.1.1, Cauchy–Schwarz inequality)

$|\langle x, y\rangle| \le \|x\|\|y\|$ for all $x, y \in \mathbb{R}^n$, with equality iff x, y are linearly dependent. Consequently $\|x + y\| \le \|x\| + \|y\|$ and $\|x - z\| \le \|x - y\| + \|y - z\|$.

Proof. If $y = 0$ both sides vanish. Otherwise, for real t, $0 \le \|x - ty\|^2 = \|x\|^2 - 2t\langle x, y\rangle + t^2\|y\|^2$; take $t = \langle x, y\rangle / \|y\|^2$ to get $0 \le \|x\|^2 - \langle x, y\rangle^2/\|y\|^2$. Equality forces $x = ty$. Then $\|x + y\|^2 = \|x\|^2 + 2\langle x, y\rangle + \|y\|^2 \le (\|x\| + \|y\|)^2$. ∎

Also $\max_i |x_i| \le \|x\| \le \sqrt{n}\max_i |x_i|$, so a sequence in $\mathbb{R}^n$ converges iff each coordinate sequence converges.

2.2 Open and closed sets

The **ϵ-neighbourhood** of x is $N(x, \epsilon) = \{y : \|y - x\| < \epsilon\}$. A point x is an **interior point** of A if some $N(x, \epsilon) \subset A$; A is **open** if all its points are interior. x is a **limit point** of A if every $N(x, \epsilon)$ contains a point of A other than x; A' denotes the set of limit points, and $\overline{A} = A \cup A'$ is the **closure**. A is **closed** if $\mathbb{R}^n - A$ is open. Points of $A - A'$ are **isolated points**. Every $N(x, \epsilon)$ is open, by the triangle inequality.

Proposition 2.2 (2.2.1)

(a) $\emptyset$ and $\mathbb{R}^n$ are open;
(b) unions of open sets are open;
(c) finite intersections of open sets are open. Dually, $\emptyset, \mathbb{R}^n$ are closed, intersections of closed sets are closed, and finite unions of closed sets are closed.

Proof. (a), (b) are immediate.

(c) If $x \in U_1 \cap \cdots \cap U_k$ with $N(x, \epsilon_i) \subset U_i$, then $N(x, \min \epsilon_i)$ lies in the intersection. The dual statements follow by De Morgan's laws. ∎

An infinite intersection of open sets need not be open: $\bigcap_n (-1/n, 1/n) = \{0\}$.

Proposition 2.3 (2.2.2)

For $A \subset \mathbb{R}^n$ and $x \in \mathbb{R}^n$:
(a) $x \in A'$ iff some sequence in $A - \{x\}$ converges to x;
(b) $x \in \overline{A}$ iff some sequence in A converges to x.

Proof. (a) If $x \in A'$, pick $x_k \in A \cap N(x, 1/k)$, $x_k \ne x$. Conversely, if $x_k \to x$ with $x_k \in A - \{x\}$, every $N(x, \epsilon)$ contains some x_k.

(b) If $x \in A$ use the constant sequence; otherwise $x \in A'$ and use (a). Conversely, if $x_k \to x$, $x_k \in A$, either $x = x_k$ for some k or the x_k lie in $A - \{x\}$. ∎

Theorem 2.4 (2.2.3)

For $F \subset \mathbb{R}^n$ the following are equivalent:
(a) F is closed;
(b) $F' \subset F$;
(c) $F = \overline{F}$.

Proof. (a)⇒(b): if $x \notin F$, some $N(x,\epsilon)$ misses F, so $x \notin F'$. (b)⇒(c): $\overline{F} = F \cup F' = F$. (c)⇒(a): if $x \notin F$, then $x \notin F'$, so some $N(x,\epsilon)$ contains no point of F (other than x, which is not in F); thus $\mathbb{R}^n - F$ is open. ■

Corollary 2.5 (2.2.4)

A' is closed for every $A \subset \mathbb{R}^n$; hence $\overline{A}$ is closed, and it is the smallest closed set containing A.

Proof. Let $x \in (A')'$ and $\epsilon > 0$. Choose $y \in A'$ with $0 < \|y - x\| < \epsilon$, and $\delta = \min(\|y - x\|, \epsilon - \|y - x\|) > 0$. Then $N(y,\delta)$ contains a point $z \in A$ with $z \neq y$; $z \in N(x,\epsilon)$, and $z \neq x$ because $\|z - y\| < \|x - y\|$. So $x \in A'$. The same argument shows $(\overline{A})' \subset A'$: if $N(x,\epsilon)$ contains $y \in A \cup A'$ with $y \neq x$, either $y \in A$, or $y \in A'$ and a small neighbourhood of y inside $N(x,\epsilon)$ contains a point of A other than x. So $\overline{A}$ is closed by Theorem 2.4. If $F \supset A$ is closed, $F \supset F' \supset A'$, so $F \supset \overline{A}$. ■

For $X \subset \mathbb{R}^n$, a set $U \subset X$ is **open in** X (relatively open) if for each $x \in U$ there is $\epsilon > 0$ with $N(x,\epsilon) \cap X \subset U$.

Proposition 2.6 (2.2.5)

For $U \subset X \subset \mathbb{R}^n$ the following are equivalent:
(a) U is open in X;
(b) $U = V \cap X$ for some open $V \subset \mathbb{R}^n$.

Proof. (a)⇒(b): for $x \in U$ choose ϵ_x with $N(x,\epsilon_x) \cap X \subset U$, and let $V = \bigcup_x N(x,\epsilon_x)$. (b)⇒(a): if $x \in V \cap X$ and $N(x,\epsilon) \subset V$, then $N(x,\epsilon) \cap X \subset U$. ■

Similarly, $F \subset X$ is closed in X iff $F = E \cap X$ with E closed in $\mathbb{R}^n$.

2.3 The Bolzano–Weierstrass theorem and Cauchy sequences

A **box** is $B = I_1 \times \cdots \times I_n$ with bounded closed intervals I_k.

Lemma 2.7 (2.3.1, nested boxes)

If $B_1 \supset B_2 \supset \cdots$ are boxes, then $\bigcap_m B_m \neq \emptyset$.

Proof. Write $B_m = I_1^m \times \cdots \times I_n^m$. For each k, the intervals I_k^m are nested, so contain a common point x_k (Theorem 1.20); then $(x_1, \ldots, x_n) \in \bigcap B_m$. ■

Theorem 2.8 (2.3.2, Bolzano–Weierstrass)

Every bounded infinite set $A \subset \mathbb{R}^n$ has a limit point.

Proof. A lies in a box B_1. Divide B_1 into 2^n boxes by bisecting each side; one of them, B_2, contains infinitely many points of A. Repeating, we obtain nested boxes B_m whose sides have length 2^{1-m}

times those of B_1, each containing infinitely many points of A. Let $x \in \bigcap B_m$ (Lemma 2.7). Any $N(x,\epsilon)$ contains B_m for large m (its diameter tends to 0), hence infinitely many points of A; so $x \in A'$. ■

Proposition 2.9 (2.3.3)

If x is a limit point of A, then $A \cap N(x,\epsilon)$ is infinite for every $\epsilon > 0$.

Proof. If $A \cap N(x,\epsilon) - \{x\} = \{a_1, \ldots, a_k\}$, then $N(x,\delta)$ with $\delta = \min\|a_i - x\|$ contains no point of A other than x, a contradiction. ■

Theorem 2.10 (2.3.4)

Every bounded sequence in $\mathbb{R}^n$ has a convergent subsequence.

Proof. If the range $\{x_k\}$ is finite, some value is taken infinitely often, giving a constant subsequence. Otherwise it is a bounded infinite set with a limit point x (Theorem 2.8). Each $N(x,1/j)$ contains infinitely many terms (Proposition 2.9), so we can choose $k_1 < k_2 < \cdots$ with $\|x_{k_j} - x\| < 1/j$. ■

For a bounded real sequence (a_n) let A be the set of limits of its convergent subsequences.

Theorem 2.11 (2.3.5)

$\limsup a_n = \max A$ and $\liminf a_n = \min A$.

Proof. Let $a = \limsup a_n$. By Proposition 1.17(b) we may choose $n_1 < n_2 < \cdots$ with $a_{n_k} > a - 1/k$, and by (a) of the same proposition $a_{n_k} < a + 1/k$ for large k; so $a_{n_k} \to a$ and $a \in A$. If $a_{n_k} \to b$, then $a_{n_k} < a + \epsilon$ for large k, so $b \le a$. The $\liminf$ case is symmetric. ■

A sequence is **Cauchy** if for every $\epsilon > 0$ there is N with $\|x_m - x_n\| < \epsilon$ for all $m, n \ge N$.

Proposition 2.12 (2.3.6)

Every convergent sequence is Cauchy, and every Cauchy sequence is bounded.

Proof. If $x_n \to x$, choose N with $\|x_n - x\| < \epsilon/2$ for $n \ge N$; then $\|x_m - x_n\| < \epsilon$. If (x_n) is Cauchy, choose N for $\epsilon = 1$; then $\|x_n\| \le \max\{\|x_1\|, \ldots, \|x_N\|\} + 1$. ■

Lemma 2.13 (2.3.7)

If a subsequence of a Cauchy sequence (x_n) converges to x, then (x_n) converges to x.

Proof. Given ϵ, choose N with $\|x_m - x_n\| < \epsilon/2$ for $m, n \ge N$, and k with $n_k \ge N$ and $\|x_{n_k} - x\| < \epsilon/2$. For $n \ge N$, $\|x_n - x\| \le \|x_n - x_{n_k}\| + \|x_{n_k} - x\| < \epsilon$. ■

Theorem 2.14 (2.3.8, Cauchy criterion)

Every Cauchy sequence in $\mathbb{R}^n$ converges.

Proof. It is bounded (Proposition 2.12), so has a convergent subsequence (Theorem 2.10); apply Lemma 2.13. ■

Corollary 2.15 (2.3.9, Cauchy criterion for series)

$\sum a_n$ converges iff for every $\epsilon > 0$ there is N with $|a_{m+1} + \cdots + a_n| < \epsilon$ for all $n > m \geq N$.

Proof. Apply Theorem 2.14 to the partial sums s_n, since $s_n - s_m = a_{m+1} + \cdots + a_n$. ■

In particular, if $\sum a_n$ converges then $a_n \to 0$, and if $\sum |a_n|$ converges (**absolute convergence**) then $\sum a_n$ converges, because $|a_{m+1} + \cdots + a_n| \leq |a_{m+1}| + \cdots + |a_n|$.

2.4 Series

Proposition 2.16 (2.4.1, rearrangement of nonnegative series)

Let $a_n \geq 0$ and $\tau : \mathbb{N} \to \mathbb{N}$ a bijection. If $\sum a_n = s$, then $\sum a_{\tau(n)} = s$; if $\sum a_n$ diverges, so does $\sum a_{\tau(n)}$.

Proof. Each partial sum $\sum_{n \leq N} a_{\tau(n)}$ is at most $\sum_{n \leq M} a_n \leq s$ with $M = \max \tau(\{1, \ldots, N\})$, so $t = \sum a_{\tau(n)} \leq s$ by Corollary 1.16. Applying this to τ^{-1} gives $s \leq t$. The divergence statement follows by symmetry. ■

More generally, an absolutely convergent series may be rearranged without changing its sum: split a_n into positive and negative parts $a_n^{\pm} = \max(\pm a_n, 0)$, both summable, and apply the proposition to each. By contrast, a conditionally convergent series can be rearranged to have any sum (Riemann).

Theorem 2.17 (2.4.2, alternating series test)

If $x_1 \geq x_2 \geq \cdots \geq 0$ and $x_n \to 0$, then $\sum (-1)^{n+1} x_n$ converges, and its sum s satisfies $|s - s_n| \leq x_{n+1}$.

Proof. $s_{2m} = (x_1 - x_2) + \cdots + (x_{2m-1} - x_{2m})$ increases, $s_{2m+1} = x_1 - (x_2 - x_3) - \ldots$ decreases, and $s_{2m} \leq s_{2m+1}$. Both converge (Theorem 1.15), and $s_{2m+1} - s_{2m} = x_{2m+1} \to 0$, so they have the same limit s, which lies between consecutive partial sums. ■

Theorem 2.18 (2.4.3, comparison test)

If $0 \leq a_n \leq b_n$ and $\sum b_n$ converges, then $\sum a_n$ converges.

Proof. The partial sums of $\sum a_n$ are bounded by $\sum b_n$; use Corollary 1.16. ■

Proposition 2.19 (2.4.4, root test)

Let $\alpha = \limsup |a_n|^{1/n}$. If $\alpha < 1$, $\sum a_n$ converges absolutely; if $\alpha > 1$, $\sum a_n$ diverges.

Proof. If $\alpha < \beta < 1$, then $|a_n| < \beta^n$ for large n (Proposition 1.17), and $\sum \beta^n < \infty$. If $\alpha > 1$, $|a_n| > 1$ for infinitely many n, so $a_n \not\to 0$. ■

Proposition 2.20 (2.4.5, ratio test)

Let $a_n \neq 0$. If $\limsup |a_{n+1}/a_n| < 1$, then $\sum a_n$ converges absolutely; if $|a_{n+1}/a_n| \geq 1$ for all large n, then $\sum a_n$ diverges.

Proof. If $|a_{n+1}/a_n| \le \beta < 1$ for $n \ge N$, then $|a_n| \le |a_N|\beta^{n-N}$. In the second case $|a_n|$ is eventually nondecreasing and positive, so $a_n \not\to 0$. ■

Proposition 2.21 (2.4.6)

If $c_n > 0$, then $\liminf \frac{c_{n+1}}{c_n} \le \liminf c_n^{1/n} \le \limsup c_n^{1/n} \le \limsup \frac{c_{n+1}}{c_n}$.

Proof. Let $\beta = \limsup c_{n+1}/c_n$, assumed finite, and $\gamma > \beta$. For $n \ge N$, $c_{n+1} < \gamma c_n$, so $c_n < c_N\gamma^{-N}\gamma^n$ and $c_n^{1/n} < (c_N\gamma^{-N})^{1/n}\gamma \to \gamma$. Hence $\limsup c_n^{1/n} \le \gamma$ for all $\gamma > \beta$. The left inequality is analogous. ■

So the root test is stronger than the ratio test: whenever the ratio test decides, so does the root test.

2.5 Compact sets

An **open cover** of K is a family of open sets whose union contains K; K is **compact** if every open cover has a finite subcover.

Proposition 2.22 (2.5.1)

Every compact set is bounded and closed.

Proof. The sets $N(0,k)$ cover K; finitely many do, so $K \subset N(0,m)$. If $x \notin K$, the open sets $U_k = \{y : \|y - x\| > 1/k\}$ cover K; a finite subcover gives $K \subset U_m$, so $N(x, 1/m)$ misses K. Hence $\mathbb{R}^n - K$ is open. ■

Theorem 2.23 (2.5.2)

Every box $B = I_1 \times \cdots \times I_n$ is compact.

Proof. Suppose an open cover $\{U_i\}$ of B has no finite subcover. Bisecting all sides, one of the 2^n sub-boxes, B_2, is not covered by finitely many U_i; continuing gives nested boxes B_m of diameter $\operatorname{diam} B_1/2^{m-1}$ none finitely covered. Let $x \in \bigcap B_m$ (Lemma 2.7) and $x \in U_i$, $N(x,\epsilon) \subset U_i$. For large m, $B_m \subset N(x,\epsilon) \subset U_i$, a contradiction. ■

Proposition 2.24 (2.5.3)

A closed subset F of a compact set K is compact.

Proof. If $\{U_i\}$ covers F, then $\{U_i\} \cup \{\mathbb{R}^n - F\}$ covers K; a finite subcover, with $\mathbb{R}^n - F$ removed, covers F. ■

Theorem 2.25 (2.5.4, Heine–Borel)

For $K \subset \mathbb{R}^n$ the following are equivalent:
(a) K is compact;
(b) K is bounded and closed;
(c) every infinite subset of K has a limit point in K.

Proof. (a)⇒(b): Proposition 2.22. (b)⇒(a): K lies in a box, which is compact (Theorem 2.23); apply Proposition 2.24. (b)⇒(c): Bolzano–Weierstrass gives a limit point, which is in K as K is

closed. (c)⇒(b): if K were unbounded, choose $x_k \in K$ with $\|x_k\| > k$; the infinite set $\{x_k\}$ has no limit point at all. If $x \in K'$, Proposition 2.9 gives distinct $x_k \in K$ with $x_k \to x$; the set $\{x_k\}$ has x as its only limit point, so $x \in K$ by (c), and K is closed. ■

So K is compact iff every sequence in K has a subsequence converging in K.

Theorem 2.26 (2.5.5, finite intersection property)

If $\{K_i : i \in I\}$ are compact sets such that $\bigcap_{i\in J} K_i \neq \emptyset$ for every finite $J \subset I$, then $\bigcap_{i\in I} K_i \neq \emptyset$.

Proof. Fix i_0. If the intersection were empty, the open sets $\mathbb{R}^n - K_i$ would cover K_{i_0}, and finitely many $\mathbb{R}^n - K_{i_1}, \dots, \mathbb{R}^n - K_{i_m}$ would cover it; then $K_{i_0} \cap K_{i_1} \cap \cdots \cap K_{i_m} = \emptyset$, a contradiction. ■

In particular a decreasing sequence of nonempty compact sets has nonempty intersection.

Corollary 2.27 (2.5.6)

A nonempty countable closed set $S \subset \mathbb{R}^n$ has an isolated point.

Proof. Let $S = \{s_1, s_2, \dots\}$ and suppose no point is isolated. Choose a closed ball $K_1 = \overline{N(y_1, r_1)}$ with $y_1 \in S$; inductively, since y_k is not isolated, $N(y_k, r_k)$ contains infinitely many points of S, so we can choose $y_{k+1} \in S \cap N(y_k, r_k)$, $y_{k+1} \neq s_k$, and a closed ball K_{k+1} about y_{k+1} inside $N(y_k, r_k)$ not containing s_k. The sets $K_k \cap S$ are nonempty, compact and decreasing, so they have a common point (Theorem 2.26); it lies in S but differs from every s_k, a contradiction. ■

Thus a nonempty closed set without isolated points (a **perfect set**), such as the Cantor set, is uncountable.

2.6 Connected sets

$S \subset \mathbb{R}^n$ is **disconnected** if there are open U, V with

$$U \cap S \neq \emptyset, \quad V \cap S \neq \emptyset, \quad U \cap V \cap S = \emptyset, \quad S \subset U \cup V,$$

and **connected** otherwise. A set $S \subset \mathbb{R}$ is an **interval** if $x, y \in S$ and $x < z < y$ imply $z \in S$.

Proposition 2.28 (2.6.1)

A subset S of $\mathbb{R}$ is connected iff it is an interval.

Proof. If S is not an interval, take $x < z < y$ with $x, y \in S$, $z \notin S$; then $U = (-\infty, z)$, $V = (z, \infty)$ disconnect S. Conversely let S be an interval and U, V as above, with $x \in U \cap S$, $y \in V \cap S$, say $x < y$; then $[x, y] \subset S$. Let $c = \sup([x, y] \cap U)$. If $c \in U$, then $c < y$ (as $y \notin U$) and U contains $[c, c+\delta)$, contradicting the definition of c; if $c \in V$, then $c > x$ and $(c-\delta, c] \subset V$, so $(c-\delta, c]$ misses U and c is not the supremum. But $c \in [x, y] \subset S \subset U \cup V$, a contradiction. ■

Proposition 2.29 (2.6.2)

If $\{C_i\}$ are connected and $\bigcap C_i \neq \emptyset$, then $C = \bigcup C_i$ is connected.

Proof. Suppose U, V disconnect C, and let $p \in \bigcap C_i$, say $p \in U$. Each C_i is connected and meets U (at p), so $C_i \cap V = \emptyset$ (otherwise U, V would disconnect C_i). Then $C \cap V = \emptyset$, a contradiction. ■

Convex sets, in particular balls and boxes, are connected: they are unions of segments through a common point, each segment being connected as a continuous image of an interval (Theorem 3.13).